SPECIFICATION FOR PRECISION SHAFT & FLANGE ASSEMBLY

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Table of Contents

1 Scope 1

2 System Requirements 1

3 Manufacturing Tolerances 4

1 Scope

This engineering specification governs the precision machining, dimensional limits, and quality inspection for heavy industrial centrifugal pump drive shafts.

2 System Requirements

The pump shaft operates in harsh environmental conditions.
The maximum operating temperature = 80°C under continuous rated load.
Verify that all maintanence schedules are strictly followed.
The assembly must withstand maximum operating pressure = 10 bar without leakage.

3 Manufacturing Tolerances

3.1 Shaft Machining Limits

Critical bearing journal diameter tolerance = ±0.5 mm on all bearing seating zones.
The specified surface roughness = 3.2 um for general unground surfaces.

3.3 Environmental Operating Limits

The drive shaft maximum operating temperature = 60°C during continuous operation.
Note that teh pump casing was recieved in good condition.

Table 1: Critical Component Dimensions

Component	Nominal Dimension	Machining Tolerance
Shaft Journal	50 mm	±0.05 mm
Flange Collar	120 mm	-
Coupling Hub	12.5 cm	±0.02 mm